Load-Adaptive and Energy-Efficient Topology Control in LEO Mega-Constellation Networks

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***Abstract*—The Low-Earth-Orbit (LEO) mega-constellation networks, by providing low-latency and high-speed communications, are becoming indispensable infrastructures for the future six-generation (6G) architecture. Consequently, the topology, with thousands of satellites equipped with batteries of limited life, has to be adaptively controlled with high energy efficiency. However, existing work lacks the joint consideration of energy efficiency and load adaptation. In this paper, we first propose the *line-of-sight condition* to determine the candidate ISL set. Next, we model the energy consumption of the LEO mega-constellation networks. Along this direction, we formulate the *Load-Adaptive and Energy-Efficient (LAEE)* topology control problem in LEO mega-constellation networks and prove its NP-hardness. Finally, we propose the *Amortized Energy based Topology Control (AETC)* algorithm to solve the LAEE problem, with good adaptation to the fluctuating load and guarantees connectivities between any two satellites. Extensive simulation results demonstrate that the AETC algorithm outperforms related schemes in terms of energy consumption and results in good topology stability.**

I. Introduction

The Low-Earth-Orbit (LEO) mega-constellation networks, with more than 10 times the number of satellites in orbit today, have been widely studied in recent years [1]. Compared with traditional constellations such as the Iridium system, the LEO mega-constellation networks are believed to be indispensable infrastructures for the six-generation (6G) and satellite based Internet of Things architecture in the foreseeable future [2], [3]. However, the limited number of complete discharge and charge battery cycles of the on-orbit satellites challenges the mega-constellation design, for example, it will be a massive device update cost for Starlink with eventually more than 40,000 nodes. Besides, complex link switches and fluctuating loads further complicate the topology control. Hence, *realizing energy-efficient topology control with good adaptation to the varying load is a critical but difficult problem* [4], [5].

Existing related research either aims to improve energy [1], [6], [7] or adapt the load variation [4], [8], [9], lacking the joint consideration of energy efficiency and load adaptation. By

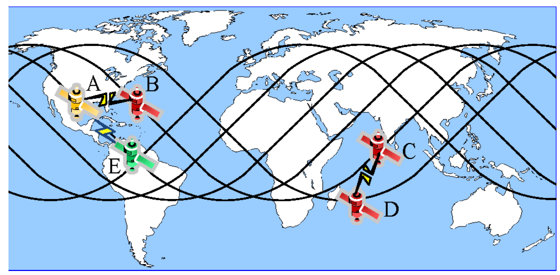

Fig. 1: Load-adaptive and energy-efficient topology control

taking advantage of steerable antennas and phased arrays, we observe that it is possible for satellites in mega-constellations to adaptively establish multiple inter-satellite links (ISLs) across different orbits based on the load level and energy consumption, guaranteeing connectivity simultaneously.

For example, as shown in Fig. 1, the ISL between satellites A and B is switched off due to the load increases of satellite B. Meanwhile, the ISL between satellites C and D is switched off considering the reduction of their load. To jointly meet the connectivity requirement and improve both the energy efficiency and the topology stability, the topology will be updated by establishing the ISL between satellites A and E at the beginning of the next slot. To adaptively construct the energy-efficient topology in the complex and dynamic LEO mega-constellation networks, two *key* problems exist as

(1) How to model the energy consumption of the topology?
(2) How to dynamically determine the topology with good adaptation to the load variation and energy efficiency?

In this paper, we investigate how to dynamically construct topology with good adaptation to the load and improve the energy consumption in LEO mega-constellation networks. To determine the candidate ISLs, we propose the line-of-sight condition. Besides, we conduct a rigorous analysis of the energy consumption in LEO mega-constellation networks. The main contributions are summarized as follows.

1) We formulate the *Load-Adaptive and Energy-Efficient (LAEE)* topology control problem in LEO mega-

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constellation networks, aiming to minimize the cost based on the line-of-sight condition and the energy consumption model; then, we prove that the LAEE problem is NP-hard.
2) We propose the *Amortized Energy based Topology Control (AETC)* algorithm, with good adaptation to the fluctuating load and guarantees connectivities between any nodes.
3) We evaluate our AETC algorithm. Comprehensive simulation results demonstrate that our AETC algorithm significantly outperforms the representative algorithms in terms of energy consumption and has good topology stability.

The remainder of this paper is organized as follows. Section II briefly reviews related work. The system model and problem formulation are presented in Section III. In Section IV, we propose the Amortized Energy based Topology Control (AETC) algorithm. Section V evaluates the AETC algorithm. We conclude this paper in Section VI.

II. Related Work

A. Energy-Improved Topology Control

In satellite networks, Ji et al [6] improved the energy of mega-constellations by jointly considering the satellite coverage factor and temporal control effectiveness. They assumed that each satellite can establish at most four ISLs including two intra-orbit ISLs and two inter-orbit ISLs. The inter-orbit ISLs were only switched off in the polar area without adaptation to the time-varying load. Authors in [7] proposed a method to encode the algebraic connectivity to reflect the robust and the energy cost of the satellite topology, where they modeled the link attenuation as a function of the distance between the transmitter and receiver telescopes. However, the energy cost of maintaining the ISL lacked consideration. Liu et al [10] designed the energy-efficient constellation topology by making full use of the limited antennas, where the topology control was modeled as the degree-bounded perfect matching problem, but it cannot well adapt to the fluctuating load.

In space-terrestrial integrated networks, OrbitCast [1] was proposed by exploiting a location-driven topology control method to adaptively choose the best ground stations. The energy consumption was greatly reduced due to the improved transmission time and traffic overhead.

B. Load-Adaptive Topology Control

In single-layer satellite networks, Lee et al [8] proposed a serviceable adjacent satellite determination method and adopted the antenna steering capability to establish ISL with other serviceable adjacent satellites, which alleviates the load of neighbors compared with the traditional LEO constellations that only contain intra- and inter-orbit ISLs. Authors in [4] conducted rigorous theoretical analysis about the relationship between the hop-count and satellite relative locations. By classifying the motion patterns of satellites into ascending and descending categories, the authors also provided a user-location-aware and load-adaptive topology control method for mega-constellations.

In multi-layer satellite networks, Li et al. [9] proposed a time-relevant graph model to reflect the cost of topology in the space and time dimensions. They aimed to reduce the congestion level of the whole network by finding the time-evolving connected dominating set in the time-relevant graph.

To conclude, our work in this paper differs from the existing work by jointly well adapting to the time-varying load and reducing the energy cost in LEO mega-constellation networks, with guaranteed end-to-end network connectivities and limited link switch overhead.

III. System Model and Problem Formulation

A. LEO Mega-Constellation Networks

The LEO mega-constellation networks consist of N orbits with the same inclination angle i, which is the angle between the orbit and the equator. Each orbit has M satellites with the same operation height H above the surface of the Earth. We assume that satellites of the same orbit are evenly distributed. Differing from the traditional LEO satellite constellation, we assume that satellites can establish inter-satellite links (ISLs) through radio-frequency communication with other satellites if they satisfy the *line-of-sight condition*.

The number of ISLs for each satellite is not restricted. In other words, there exists a candidate ISL if the line-of-sight condition is met. We adopt the software-defined architecture proposed in [11] to manage our LEO mega-constellation networks. Specifically, the period T is split into a series of slots with equal length τ. The topology connectivity of each slot is stable. At the beginning of each slot, both the controller and the super-controller jointly take the load distribution and energy consumption to construct the relatively stable and energy-efficient topology. We use $G_\tau = (V_\tau, E_\tau)$ to model the topology in slot t, where V_τ is the set of satellites and E_τ is the set of ISLs. Due to the space limitation, we focus on the topology control algorithm design and implementation instead of how to deploy it in the software-defined architecture.

Before formulating the problem, we first propose the line-of-sight condition to construct the candidate ISL set and then quantify the energy consumption of the dynamic topology.

B. Line-of-Sight Condition

The line-of-sight condition determines whether the ISL can be established between any two satellites. We put the topology of the LEO mega-constellation into a Cartesian three-dimensional coordinate system, with the center of the earth as the origin of the coordinates. Given two satellites $v, u \in V_\tau$, we denote their coordinates as (x_v, y_v, z_v) and (x_u, y_u, z_u), respectively. The line-of-sight condition holds when both of the following sub-conditions are satisfied

$$\frac{\sqrt{x'^2 + y'^2 + z'^2}}{\sqrt{(x_u - x_v)^2 + (y_u - y_v)^2 + (z_u - z_v)^2}} > R_E + D_{\min}, \quad (1)$$

$$\arcsin \frac{\max\{d_\perp^{uv}, d_\perp^{vu}\}}{\sqrt{(x_u - x_v)^2 + (y_u - y_v)^2 + (z_u - z_v)^2}} < \overline{\alpha}, \quad (2)$$

where R_E is the radius of the Earth. $d_\perp^{uv}$ (resp. $d_\perp^{vu}$) is the distance from satellite u (resp. v) to the extension line of a certain antenna of satellite v (resp. u). Three location-related parameters, x', y' and z', are derived as below

$$\begin{cases} x' = (y_u - y_v)z_u - (z_u - z_v)y_u, \\ y' = (z_u - z_v)x_u - (x_u - x_v)z_u, \\ z' = (x_u - x_v)y_u - (y_u - y_v)x_u, \end{cases} \tag{3}$$

The first sub-condition requires that the closest distance of the ISL to the Earth's surface is above the threshold $D_{\min}$. The second condition requires any two satellites to be within the cone angle $2\overline{\alpha}$ with the opposite satellite as the apex of the cone. Both of the sub-conditions can be derived based on some intuitive algebra operations. We give an example in Fig. 2 to explain the two sub-conditions. The green satellite can establish the ISL with the yellow one since the conditions (1) and (2) are satisfied, while the red satellite cannot.

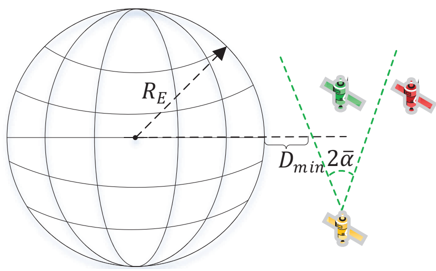

Fig. 2: Illustration of the line-of-sight condition

Based on the line-of-sight condition, we construct the candidate ISL set $\overline{E_\tau}$. Specifically, given a satellite $v \in V_\tau$, if the satellite $u \in V_\tau - \{v\}$ meets the line-of-sight conditions, we put the ISL associated with them into $\overline{E}$. Otherwise, we continue to consider another satellite until all satellites in $V_\tau - \{v\}$ have been traversed. We repeat the above process until all elements in V_τ are considered.

C. Energy Consumption Model

The energy consumption ϵ_τ during the topology control consists of three parts as below

$$\epsilon_\tau = \epsilon_\tau^{\text{on}} + \epsilon_\tau^{\text{off}} + \epsilon_\tau^{\text{ISL}}, \tag{4}$$

where $\epsilon_\tau^{\text{on}}$ and $\epsilon_\tau^{\text{off}}$ are the energy consumption of establishing and switching off the ISL, respectively. We assume that each satellite has the same energy cost to establish or switch off an ISL. We use $\epsilon_\tau^{\text{ISL}}$ to represent the ISL maintenance energy consumption. Based on [12], the $\epsilon_\tau^{\text{ISL}}$ is modeled as a linear function of the propagation delay of the ISL.

We formally define the above three types of energy consumption as below

$$\epsilon_\tau^{\text{on}} = \sum\nolimits_{e \in \overline{E_\tau}} k_1 \theta_\tau^e, \tag{5}$$

$$\epsilon_\tau^{\text{off}} = \sum\nolimits_{e \in \overline{E_\tau}} k_2 \gamma_\tau^e, \tag{6}$$

$$\epsilon_\tau^{\text{ISL}} = \sum\nolimits_{e \in \overline{E_\tau}} k_3 d_\tau^{vu} \theta_\tau^e, \tag{7}$$

$$\theta_\tau^e \in \{0,1\}, \gamma_\tau^e \in \{0,1\}, \tag{8}$$

where k_1, k_2 and k_3 are three antenna-related parameters specified by the hardware configurations. The propagation delay between satellite v and u is denoted as d_τ^{vu}. We use θ_τ^e as a binary indicator to denote whether $e \in E_\tau (\theta_\tau^e = 1)$ or not $(\theta_\tau^e = 0)$. Similarly, the γ_τ^e is another binary indicator to represent whether the ISL exists in the previous slot τ' and is switched off in this slot. Formally, we have

$$\gamma_\tau^e = \begin{cases} 1, & e \in E_{\tau'} \wedge e \notin E_\tau; \\ 0, & o.w., \end{cases} \tag{9}$$

D. Problem Formulation

Given the line-of-sight condition and the energy consumption model introduced above, we aim to minimize the cost as shown in the Formula (10) by determining the selection of ISL $(\theta_\tau^e, \gamma_\tau^e, \forall e \in \overline{E_\tau})$ in each slot τ. The *Load-Adaptive and Energy-Efficient (LAEE)* topology control problem is formulated as follows.

$$\textbf{minimize} \quad \sum\nolimits_{e \in \overline{E_\tau}} (k_1 \theta_\tau^e + k_2 \gamma_\tau^e + k_3 d_\tau^{vu} \theta_\tau^e) d_\tau^e \tag{10}$$

$$s.t. \quad (1) - (9),$$

$$d_\tau^e = d_\tau^{vu} + q_\tau^v + q_\tau^u, \forall \tau \in T, e \in \overline{E_\tau}, v, u \in V_\tau, \tag{11}$$

$$v \rightsquigarrow u, \forall v, u \in V_\tau, \tag{12}$$

where the Formula (11) is the sum of the propagation delay and queuing delays of nodes v and u, which we denote as q_τ^v and q_τ^u, respectively. The queuing delay of nodes can be easily gotten under the software-defined architecture and precisely capture the load variation. We use the notation $\rightsquigarrow$ to represent there exists a path from one node to another node. Hence, The constraint (12) ensures that the constructed topology is full-connected. Finally, to reduce energy consumption while incurring limited increases in the end-to-end delay, we take minimization of the sum of the product of energy consumption and delay as the optimization goal in the Formula (10). To sum up, the LAEE problem aims to find an energy-efficient topology from the candidate ISL set $\overline{E_\tau}$ with good adaptation to the time-varying loads in each slot.

Theorem 1. *The LAEE problem is NP-hard.*

Proof. We simplify the LAEE problem by ignoring the energy consumption of establish and switching off the ISL ($\epsilon_\tau^{\text{on}}$ and $\epsilon_\tau^{\text{off}}$). First, based on the line-of-sight condition defined in the Formula (1) and (2), we know that the number of satellites for a given satellite to establish an ISL is limited, which means the degree of the node in $G_\tau = (V_\tau, E_\tau)$ is upper bounded. Then, we treat the cost $k_3 d_\tau^{vu} \theta_\tau^e d_\tau^e$ as the link weight in G_τ. The simplified LAEE problem is equivalent to finding the degreed constrained minimum spanning tree, which is NP-hard [13]. This reduction is done within polynomial time, which proves that the original LAEE problem is NP-hard. □

IV. Amortized Energy based Topology Control

To solve the LAEE topology control problem in polynomial time, we propose the Amortized Energy based Topology Control (AETC) algorithm in this section. We first analyze that the energy cost is affected by the topology in both the current and previous slots. Then, we propose the amortized energy cost model, which amortizes the energy cost generated by switching off the ISLs in the previous slot to the cost of the newly activated ISLs associated with the same node in this slot. This model reduces the topology changes among different slots and improves stability, by pessimistically estimating the cost of switching links. Finally, we propose the Amortized Energy based Topology Control (AETC) algorithm based on the amortized energy cost model.

Algorithm 1 Amortized energy based topology control

Input: $E_{\tau'}, \{q_\tau^v, d_\tau^{vu} | v, u \in V_\tau\}$
Output: E_τ
1: **for** $v \in V_\tau$ **do**
2: Construct $\overline{E_\tau}$ based on the Formula (1) and (2)
3: **end for**
4: $\Upsilon = \emptyset, C = \text{Map} < v, cost >$
5: Randomly get a node v from V_τ
6: $\Upsilon = \Upsilon \cup \{v\}, C[v] = 0$
7: $C[u] = \infty, \forall u \in V_\tau - v$
8: **for** $u \in V_\tau$ **do**
9: **if** $\exists e_{vu} \in \overline{E_\tau}$ **then**
10: $C[u] = (k_1 \theta_\tau^{e_{vu}} + \Lambda_\tau^u + k_3 d_\tau^{vu} \theta_\tau^{e_{vu}}) d_\tau^{e_{vu}}$
11: **end if**
12: **end for**
13: **while** $\exists v, v' \in V_\tau, v \neg \rightsquigarrow v'$ **do**
14: $u = \arg\min_{u \notin \Upsilon} C[u]$
15: $\Upsilon = \Upsilon \cup \{u\}$
16: $E_\tau = E_\tau \cup \{e_{vu}\}$
17: **for** $u' \in V_\tau \wedge u' \notin \Upsilon$ **do**
18: **if** $(k_1 \theta_\tau^{e_{uu'}} + \Lambda_\tau^{u'} + k_3 d_\tau^{uu'} \theta_\tau^{e_{uu'}}) d_\tau^{e_{uu'}} < C[u']$ **then**
19: $C[u'] = (k_1 \theta_\tau^{e_{uu'}} + \Lambda_\tau^{u'} + k_3 d_\tau^{uu'} \theta_\tau^{e_{uu'}}) d_\tau^{e_{uu'}}$
20: **end if**
21: **end for**
22: **end while**

Given any satellite $v \in V_\tau$, a set of ISLs associated with v exists in the previous slot τ' may be switched off in the current slot τ. Meanwhile, a set of ISL associated with v will be activated in this slot. During the construction of the new topology, it is necessary to quantify the cost of switching off ISLs in the previous topology. An ideal way is to amortize the ISL switching off energy cost evenly to the cost of all new ISLs. However, the topology control algorithm usually starts from a single node or an edge, which means it cannot amortize the cost evenly until the final topology is built.

To approximately quantify the energy cost of switching off ISLs, we propose the amortized energy cost model. Specifically, given a satellite v, denote the set of ISLs associated with v as E_τ^v. Then, the amortized cost of adding a new ISL associated with v in the current slot is the sum of the energy cost incurred by ISL switching off of all ISLs in $E_{\tau'}^v$, where τ' is the previous slot of τ. Formally, we have

$$\Lambda_\tau^v = \sum\nolimits_{e \in E_{\tau'}^v} k_2. \tag{13}$$

Based on the amortized energy cost model, we propose the amortized energy based topology control algorithm. We denote e_{vu} as the ISL from the satellite v to u. We use $v \neg \rightsquigarrow v'$ to represent that v cannot find a path with v' as destination.

As shown in Algorithm 1, the AETC algorithm takes $E_{\tau'}$ and $\{q_\tau^v, d_\tau^{vu} | v, u \in V_\tau\}$ as inputs. Firstly, we construct the candidate ISL set $\overline{E_\tau}$ based on the line-of-sight condition defined by the Formula (1) and (2) in lines 1-3. Then, we define an empty set Υ and a map-like data structure C, which takes the node v and cost as its key and value, respectively. The cost of the node v (resp. other nodes) is initialized as 0 (resp. infinity) in lines 6-7.

We begin to construct the topology by randomly get a node v from V_τ. Next, from line 8 to line 10, if there exists a candidate ISL, we update the cost of u in C based on the load situation, d_τ^{vu} , and the amortized energy cost model Λ_τ^u. From line line 14 to line line 16, we find the node u that has the minimum cost and is not put into Υ. The ISL from satellite v and u is added into E_τ. Finally, we update the C if the cost from u to u' is smaller than $C[u']$. The above process is repeated until the topology is full-connected, which indicates that the constraint (12) is satisfied.

V. Performance Evaluations

A. System Setup

We built a simulation system based on NS-3 [14]. We evaluate our proposed AETC algorithm on two typical LEO mega-constellations, which are Starlink (Phase I) [15] and OneWeb [16]. Key parameters are listed in TABLE I.

TABLE I: MAIN NOTATIONS

Parameters	Mega-constellations	
	Starlink (Phase I)	OneWeb
Height	550km	1200km
Inclination	53°	87.9°
# of orbits	24	18
# of satellites per orbit	66	40

For the AETC algorithm, we set the threshold $D_{\min}$ of the line-of-sight condition as 200 kilometers. We set $k_3 = 1$ to indicate that the energy to maintain an ISL is linear to d_τ^{vu}. We assume the energy cost to establish and switch off an ISL is the same and set $k_1 = k_2 = \min_{v,u \in V_\tau} d_\tau^{vu}$, which are determined by the topology features of the specific mega-constellation. We compare the AETC algorithm against two related schemes:

- *OSN* [6] establishes at most four ISLs for each satellites. Two of them are intra-orbit ISLs, existing during the whole period. Another two of them are inter-orbit ISLs, which are switched off when one node of them enters the polar area. We set the latitude threshold for Starlink and OneWeb as ±50° and ±88°, respectively, where the negative latitude indicates the southern hemisphere.
- *SAS* [8] make no restriction on the number of ISL of a satellite if the visible conditions are met. In detail, it assumes that if the ISL between two satellites is not blocked by the Earth, they are regarded as the serviceable adjacent satellites and the ISL is established.

TABLE II: The logarithm of the total cost with base 2 in Starlink

Algorithms \ Load factor	1	1.5	2	2.5	3	3.5	4	4.5	5
AETC	10.488141	10.532638	10.596888	10.65839	10.717385	10.774062	10.828596	10.881144	10.931838
OSN	18.235364	18.351474	18.458931	18.558936	18.652459	18.740284	18.823068	18.901361	18.975621
SAS	20.026128	20.027857	20.02957	20.031294	20.033016	20.034723	20.036441	20.036683	20.039872

TABLE III: The logarithm of the total cost with base 2 in OneWeb

Algorithms \ Load factor	1	1.5	2	2.5	3	3.5	4	4.5	5
AETC	10.324866	10.406321	10.496713	10.581765	10.662089	10.738168	10.81044	10.879261	10.944953
OSN	17.424428	17.551362	17.668033	17.775963	17.876385	17.970263	18.058404	18.141473	18.220014
SAS	20.282993	20.28497	20.286945	20.288929	20.290898	20.292865	20.294829	20.29679	20.29876

All experiments are simulated for 3000 seconds and we take the average results. We split the period of mega-constellations into a series of one-second slots. All algorithms update the topology at the beginning of each slot. For brevity, we use Starlink to refer Starlink (Phase I) thereafter.

B. Results and Analysis

1) Energy consumption: This set of experiments observes the energy consumption under different loads. To clearly reflect the gap between our AETC algorithm and other schemes, we take the base 2 logarithms of the total cost and normalized the results of the OSN and the SAS to the AETC algorithm. We multiply the number of requests by the load factor, which is a real number, to indicate the overall load level of the LEO mega-constellation networks.

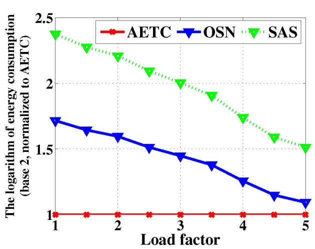

Fig. 3: Energy consumption with load in Starlink

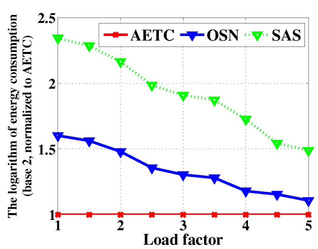

Fig. 4: Energy consumption with load in OneWeb

As shown in Fig. 3 and Fig. 4, the normalized energy cost of both the OSN and the SAS are higher than the AETC algorithm. On the one hand, the SAS always selects all visible satellites to establish ISLs, incurring the highest energy consumption. On the other hand, the intra-orbit ISLs under the OSN scheme always exist during the whole period and the inter-orbit ISLs are only switched off in the polar area, lacking the adaptation to the variation of the load.

With the increase of load, the energy consumption of the OSN and the SAS decreases, which indicates that the topology constructed by the AETC algorithm increases, which shows that our AETC dynamically updates the topology based on the load level of the whole network, at the beginning of each slot. In other words, the increase in energy consumption of the AETC results from choosing other ISLs to minimize the total cost due to the increase in load.

2) Total cost: We compare the performance of AETC with OSN and SAS in terms of the total cost. We take the logarithm of the total cost with base 2. As shown in TABLE II and TABLE III, our AETC has the smallest total cost under different load levels. The reason is that the AETC selects a subset of ISLs from the $\overline{E_\tau}$ with the goal of minimizing the total cost. On the contrary, the SAS has the largest total cost since it greedily establishes ISLs for each satellite with other visible nodes. Although the OSN aims to improve energy consumption by switching off inter-orbit ISLs based on the locations of satellites, there still exist redundant ISLs which incur extra energy consumption.

Further, the increase of total cost with the increase of load results from two reasons. The first is the overall increase in the queuing delay of all satellites. The second is our AETC algorithm chooses another ISL with higher energy consumption to guarantee connectivity. However, the AETC has a smaller relative increase than the other two schemes, which shows the AETC's good adaption to the load variation.

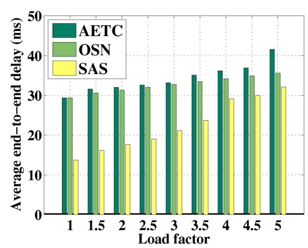

Fig. 5: Avg. e2e delay (Starlink)

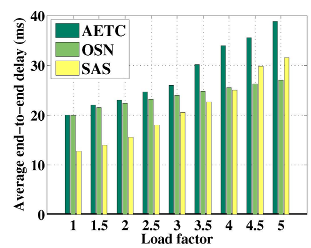

Fig. 6: Avg. e2e delay (OneWeb)

3) Average end-to-end delay: We select 1000 pairs of sources and destinations from the LEO mega-constellation networks and evaluate the average end-to-end (e2e) delay, which is a potential overhead incurred by our energy-efficient topology control algorithm. The path is constructed based on the shortest path algorithm with propagation and queuing delay as routing metrics. As shown in Fig. 5 and 6, the end-to-end delay of our AETC is relatively higher than the other two schemes, whereas, the SAS has the lowest end-to-end delay.

These results attribute to two reasons. Firstly, our AETC aims to minimize the total cost of the topology, which is the sum of the product of delay and energy consumption. As

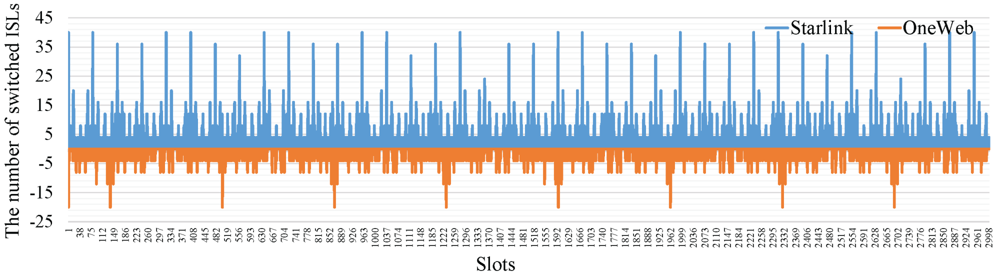

Fig. 7: The number of switched ISLs in Starlink and OneWeb

a result, some of the candidate ISLs in $\overline{E_\tau}$, which belong to the optimal path, may not be selected in the constructed topology. Secondly, satellites under the SAS scheme establish ISLs with all ISLs, which decreases the number of hops along the paths. With the increase in load, the overall queuing delay of all satellites grows. However, the reduction in the number of hops makes the end-to-end delay less affected by the increase in load. In the OneWeb constellation, the SAS has a higher average end-to-end delay than the OSN when the load factors are 4.5 and 5, which results from the fact that paths have fewer hops in OneWeb than Starlink.

4) The number of switched ISLs: We define the ISL as switched if one of the following conditions satisfies: 1) The ISL exists in the previous slot and is switched off in the current slot; 2) The ISL does not exist in the previous slot and is established in the current slot. We evaluate the total number of switched ISLs in Starlink and OneWeb to show the topology stability under our AETC algorithm. As shown in Fig. 7, the number of switched ISLs is bounded by 40 (resp. 20) in Starlink (resp. OneWeb), where the negative number corresponds to the situation in OneWeb.

Considering the huge number of ISLs in LEO mega-constellation networks such as Starlink and OneWeb, our AETC algorithm does not cause large amounts of ISL switches and results in good topology stability.

VI. Conclusion

This paper investigates the topology control problem in LEO mega-constellation networks. First, we formulate the *Load-Adaptive and Energy-Efficient (LAEE)* topology control problem, with the aim of minimizing the cost, which precisely captures the time-varying load and energy consumption. Then, to solve the LAEE problem in polynomial time, we propose the *Amortized Energy based Topology Control (AETC)* algorithm which well adapts to the fluctuating load and ensures that there exists a path between any two satellites. Extensive simulation results further demonstrate that our AETC algorithm outperforms other related schemes in terms of energy consumption and has good topology stability.

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